Muhammad Mostafa Saadawy Amin

Junior Data Engineer

elsameenm@gmail.com | +20 12 26867568 | LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Results-driven Junior Data Engineer (3rd-year student, expected graduation 2027) skilled in building scalable ETL pipelines using Python, Spark, Kafka, and Airflow. Proven experience designing Star Schema warehouses, optimizing query performance by 30–40%, and deploying cloud solutions on Microsoft Azure. Passionate about real-time data processing and CI/CD automation at scale.

EDUCATION

Bachelor of Computers and Artificial Intelligence *Expected: 2027*
Minia National University, Egypt
Relevant Coursework: Data Structures, Data Mining, Algorithms, Database Systems, OOP, Machine Learning

PROFESSIONAL EXPERIENCE

Junior Data Engineer – Trainee (DEPI Scholarship) | Ministry of Communications – DEPI Program *Jun 2025 – Dec 2025*
Completed.

- Built and maintained 10+ end-to-end batch and streaming ETL pipelines using Python, SQL, Apache Spark, Apache Kafka, and Apache Airflow for data ingestion and transformation.
- Designed 5+ data warehouse schemas (Star Schema) and optimized analytical queries, improving query performance by 30–40%.
- Processed and analyzed millions of records using Hadoop, Hive, and Spark, enabling scalable and efficient data processing workflows.
- Implemented cloud-based analytics solutions on Azure Synapse Analytics, Dedicated SQL Pools, and Azure Stream Analytics, supporting near real-time data processing.
- Applied CI/CD pipelines to automate deployment, orchestration, and monitoring using Git and GitHub Actions, reducing manual intervention by 40%.
- Worked with Azure Data Factory for data integration and pipeline orchestration across cloud data sources.

Machine Learning Trainee | National Telecommunication Institute (NTI) *Aug 2025 – Sep 2025 Completed.*

- Developed and evaluated machine learning models using Python, NumPy, Pandas, and Scikit-learn for real-world datasets.
- Applied data preprocessing, feature engineering, and model tuning to improve model performance and accuracy.
- Implemented supervised and unsupervised learning algorithms including regression, classification, and clustering.
- Built end-to-end ML workflows from data collection and cleaning to model training, evaluation, and deployment via REST APIs using Streamlit.

PROJECTS

IKEA Shopping – Admin Panel + Data Engineering & ML Pipeline GitHub | Live Demo *Apr 2026 – Jun 2026*

- Extended a Java Swing admin panel (SQL Server, 10-table normalized schema, RBAC, SHA-256 hashing) with a full Data Engineering and ML layer, without modifying any existing Java code.
- Built a synthetic data generator producing 18 months of realistic sales data with monthly seasonality, weekly patterns, and growth trend using NumPy and pyodbc.
- Designed and implemented an ETL pipeline (Extract from SQL Server → Transform into a Star Schema → Load into SQLite warehouse) with Fact_Sales, Dim_Product, and Dim_Date tables.
- Trained an XGBoost forecasting model with lag features (lag-1, lag-7), 7-day rolling mean, and chronological train/test split; deployed a live Streamlit dashboard showing actual vs. forecasted daily sales per product.

Technologies: *Java, Swing, SQL Server, Python, Pandas, NumPy, XGBoost, Streamlit, SQLite, ETL, Data Warehousing*

Smart City – Real-Time Data Pipeline GitHub *Dec 2025 – Apr 2026*

- Built an end-to-end real-time data pipeline simulating 5 smart city IoT sources (vehicles, GPS, traffic cameras, weather, emergencies) using Python producers and Apache Kafka.
- Processed concurrent Kafka streams with Apache Spark Structured Streaming, applying per-topic schemas, JSON deserialization, and watermarking to handle late-arriving events.
- Containerized the full distributed system (Zookeeper, Kafka broker, Spark master + 2 workers) using Docker Compose for one-command, reproducible deployment.
- Modeled data entities and relationships via ERD across 5 interconnected tables to support downstream analytics.

Technologies: *Python, Apache Kafka, Apache Spark Structured Streaming, Docker, Docker Compose, Apache Parquet*

Road Collisions Severity Prediction – ML Project GitHub | Live Demo *Sep 2025 – Nov 2025*

- Built an end-to-end ML system to predict road collision severity (92.6% accuracy) using real-world traffic accident data.
- Performed data cleaning, feature engineering, and EDA to identify key factors affecting accident severity.
- Trained and evaluated a CatBoost classifier, achieving 0.92 weighted F1-score with perfect precision on Fatal and Serious collision classes.
- Deployed an interactive Streamlit web application for real-time severity predictions.

Technologies: *Python, CatBoost, Scikit-learn, Pandas, Streamlit, Matplotlib, Seaborn*

Spam Email Classifier – NLP Project GitHub | Live Demo *Aug 2025*

- Developed an end-to-end NLP pipeline for spam detection using tokenization, stopword removal, lemmatization, and TF-IDF vectorization.
- Trained and evaluated classification models using Scikit-learn; managed model persistence with Joblib.
- Deployed an interactive Streamlit dashboard for real-time predictions with probability breakdown.

Technologies: *Python, NLTK, Scikit-learn, TF-IDF, Pandas, Streamlit, Git, GitHub*

TECHNICAL SKILLS

Languages	Python, SQL, C++, Java, C#
Data Engineering	ETL Pipelines, Data Modeling, Data Warehousing, Star Schema, Batch & Streaming Processing
Big Data	Apache Spark, PySpark, Apache Kafka, Hadoop, Hive, Spark Structured Streaming
Orchestration	Apache Airflow, CI/CD Automation, GitHub Actions, dbt
Cloud (Azure)	Azure Synapse Analytics, Azure Data Factory, Azure Databricks, Dedicated SQL Pools, Azure Stream Analytics
Databases	PostgreSQL, MySQL, Delta Lake, MongoDB (NoSQL)
DevOps & Tools	Docker, Docker Compose, Git, GitHub, REST APIs, Pipeline Monitoring
ML Libraries	Scikit-learn, Pandas, NumPy, CatBoost, Matplotlib, Seaborn, NLTK, Streamlit
Languages (Human)	Arabic (Native), English (Professional Working Proficiency)

LEADERSHIP & ACTIVITIES

- Mentor at ICPC MNU Team – guided competitive programming students in problem-solving and algorithms.
- Mentor at MENTOR Network – supported junior students in career development and technical growth.
- Admin at Student Activity – managed organizational and academic community events.